Lecture 2. Basic Quantities

Survival, Hazard, Cumulative Hazard, Quantiles, Residual Life

Contents

1 Review: Classifying Incomplete Data 1

2 Survival Function 1

3 Hazard Function 2

3.1 Shapes of hazard functions 2

3.2 Examples 2

4 Cumulative Hazard and the S–h Relationship 3

5 Discrete Failure Times 3

6 Mean and Mean Residual Life 4

7 Quantiles 5

8 Median Residual Life 6

9 Weibull Mean Residual Life 6

10 mrl and Hazard Monotonicity 7

11 Selected exercises 7

1 Review: Classifying Incomplete Data

Situation	Type
Trial from randomization; no event by month 24 at study end.	RC
Tumour present at baseline; onset only known to precede it.	LC
Annual screening; disease-free at year 2, first positive at year 3.	IC
Registry enrols only patients alive at clinic entry (up to 3 years after diagnosis); followed to death or study end.	LT + (RC or event)
Cohort built in 2020 from obituaries for 2010–2019; age at death only.	RT

2 Survival Function

Let $X \geq 0$ be a random failure time.

Definition 2.1: Survival function

For continuous X with pdf f and cdf $F(x) = P(X \le x)$,

$$S(x) = P(X > x) = 1 - F(x) = \int_x^\infty f(s)\,\mathrm{d}s.$$

the probability of surviving beyond x; non-increasing, $S(0) = 1$, $S(\infty) = 0$.

$S(x)$ is the area under the density right of x. (E.g. 1989 US survival curves stay near 1 until middle age, then drop steeply.)

3 Hazard Function

Definition 2.2: Hazard function

$$h(x) = \lim_{\Delta x \to 0} \frac{P(x \le X < x + \Delta x \mid X \ge x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{P(x \le X < x + \Delta x)/P(X \ge x)}{\Delta x} = \frac{f(x)}{S(x)}.$$

- $f(x)\,\mathrm{d}x \approx S(x) - S(x + \mathrm{d}x)$: failure probability at x, *unconditionally*.
- $h(x)\,\mathrm{d}x$: failure probability just after x, *given* alive at x.

Remark 2.3: h is a rate, not a probability

$h(x)$ can exceed 1; only $h(x)\Delta x$ approximates a conditional probability.

3.1 Shapes of hazard functions

(i) **Constant**: exponential.

(ii) **Increasing**: ageing industrial systems and cars.

(iii) **Decreasing**: e.g. transplants (risk highest after the operation).

(iv) **Bathtub**: population mortality (infancy high, mid-life low, old age rising).

(v) **Hump-shaped**: e.g. breast cancer treatment, where high-risk patients die early.

3.2 Examples

Example 2.4: Exponential(λ)

$f(x) = \lambda e^{-\lambda x}$ and

$$S(x) = \int_x^\infty \lambda e^{-\lambda s}\,\mathrm{d}s = \big[-e^{-\lambda s}\big]_x^\infty = e^{-\lambda x}, \qquad h(x) = \frac{f(x)}{S(x)} = \lambda.$$

Constant hazard = memorylessness: $P(X > x + t \mid X > x) = e^{-\lambda t}$.

Example 2.5: Weibull(α, λ)

$S(x) = \exp(-\lambda x^{\alpha})$. Then

$$f(x) = -\frac{\mathrm{d}S(x)}{\mathrm{d}x} = \alpha\lambda x^{\alpha-1}\exp(-\lambda x^{\alpha}), \qquad h(x) = \frac{f(x)}{S(x)} = \alpha\lambda x^{\alpha-1}.$$

Increasing for $\alpha > 1$, constant for $\alpha = 1$, decreasing for $\alpha < 1$.

4 Cumulative Hazard and the S–h Relationship

Motivation. S, f, h, H describe the same law; $S = \exp\{-H\}$ moves between them.

Theorem 2.6: Continuous case

Define the **cumulative hazard** $H(x) = \int_0^x h(s)\,\mathrm{d}s$. If X is absolutely continuous,

$$H(x) = -\ln S(x), \qquad S(x) = \exp\{-H(x)\} = \exp\Big\{-\int_0^x h(s)\,\mathrm{d}s\Big\}.$$

Proof [Supplement]. Since $f(s) = -S'(s)$,

$$H(x) = \int_0^x \frac{f(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \frac{S'(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \mathrm{d}\{\ln S(s)\} = -\big[\ln S(x) - \ln S(0)\big] = -\ln S(x),$$

using $S(0) = 1$. ∎

Summary of relations (continuous X)	
$f(t) = \dfrac{\mathrm{d}F(t)}{\mathrm{d}t} = -S'(t)$	$F(t) = 1 - S(t)$
$h(t) = \dfrac{f(t)}{S(t)} = -\dfrac{\mathrm{d}}{\mathrm{d}t}\ln S(t)$	$H(t) = \displaystyle\int_0^t h(s)\,\mathrm{d}s = -\ln S(t)$
$S(t) = \exp\{-H(t)\}$	$f(t) = h(t)\exp\{-H(t)\}$

Any one of f, F, S, h, H determines the others.

5 Discrete Failure Times

At $0 = x_0 < x_1 < \cdots < x_d$ let $P(X = x_j) = p_j > 0$, $\sum p_j = 1$ (rounding or grouping). Then

$$S(x) = P(X > x) = \sum_{j=1}^{d} P(X = x_j)\, I(x_j > x),$$

a right-continuous step function.

Definition 2.7: Discrete hazard

$$h(x_j) = P(X = x_j \mid X \geq x_j) = \frac{p_j}{S(x_{j-1})} = \frac{p_j}{\sum_{k=j}^{d} p_k}, \qquad j = 1, \ldots, d.$$

Here $h(x_j)$ *is* a conditional probability: $0 < h(x_j) \le 1$.

Since $p_j = S(x_{j-1}) - S(x_j)$,

$$h(x_j) = \frac{S(x_{j-1}) - S(x_j)}{S(x_{j-1})} = 1 - \frac{S(x_j)}{S(x_{j-1})}.$$

Example 2.8: $p_j = 1/3$, $j = 1, 2, 3$

$$S(x) = \begin{cases} 1, & 0 \le x < 1 \\ 2/3, & 1 \le x < 2 \\ 1/3, & 2 \le x < 3 \\ 0, & x \ge 3 \end{cases} \qquad h(1) = \frac{1/3}{1} = \frac{1}{3}, \quad h(2) = \frac{1/3}{2/3} = \frac{1}{2}, \quad h(3) = \frac{1/3}{1/3} = 1.$$

The last hazard is 1: survivors to the last support point fail there.

Theorem 2.9: Product-limit representation

$$S(x_j) = \frac{S(x_j)}{S(x_{j-1})}\frac{S(x_{j-1})}{S(x_{j-2})} \cdots \frac{S(x_1)}{S(x_0)} S(x_0) = \prod_{i:x_i \le x_j} \frac{S(x_i)}{S(x_{i-1})} = \prod_{i:x_i \le x_j} [1 - h(x_i)],$$

(telescoping, $S(x_0) = 1$).

This is the population Kaplan–Meier form (Lecture 5).

Remark 2.10: Two definitions of the discrete cumulative hazard

K&M define $H(x_j) = \sum_{i:x_i \le x_j} h(x_i)$, for which $S = e^{-H}$ fails. Cox & Oakes (1984) define

$$H(x_j) = -\sum_{i:x_i \le x_j} \log\left[1 - h(x_i)\right] \approx \sum_{i:x_i \le x_j} h(x_i),$$

which restores $S = e^{-H}$ exactly ($-\log(1-u) \approx u$ for small u).

6 Mean and Mean Residual Life

Motivation. The mean residual life is the expected remaining lifetime of a survivor, computable from S.

Theorem 2.11: Mean as area under S

For $X \ge 0$, $\mathrm{E}(X) = \int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty S(t)\,\mathrm{d}t$.

Proof [Supplement]. Write $t = \int_0^t \mathrm{d}u$ and use Tonelli:

$$\int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty \int_0^t f(t)\,\mathrm{d}u\,\mathrm{d}t = \int_0^\infty \int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_0^\infty S(u)\,\mathrm{d}u.$$

■

Definition 2.12: Mean residual life

$\mathrm{mrl}(x) = \mathrm{E}(X - x \mid X > x)$. For continuous X,

$$\mathrm{mrl}(x) = \frac{\int_x^\infty (t-x) f(t)\,\mathrm{d}t}{S(x)} = \frac{\int_x^\infty S(t)\,\mathrm{d}t}{S(x)},$$

the area under S right of x divided by $S(x)$; $\mathrm{mrl}(0) = \mathrm{E}(X)$.

Proof of the second equality [Supplement]. With $t - x = \int_x^t \mathrm{d}u$ and Fubini, $\int_x^\infty (t-x) f(t)\,\mathrm{d}t = \int_x^\infty \int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_x^\infty S(u)\,\mathrm{d}u$. ■

7 Quantiles

Definition 2.13: p-th quantile

The p-th quantile ($100p$-th percentile) is

$$x_p = \inf\{t : S(t) \le 1 - p\}.$$

For continuous strictly decreasing S, $S(x_p) = 1 - p$; the median is $x_{0.5}$.

Example 2.14: Exponential(λ) — mean, mrl, median

$S(x) = e^{-\lambda x}$.

- $\mathrm{E}(X) = \int_0^\infty e^{-\lambda t}\,\mathrm{d}t = 1/\lambda$.
- $\mathrm{mrl}(x) = \dfrac{e^{-\lambda x}/\lambda}{e^{-\lambda x}} = \dfrac{1}{\lambda}$ (memorylessness).
- Median: $e^{-\lambda x_{0.5}} = 1/2 \Rightarrow x_{0.5} = (\ln 2)/\lambda$.

Example 2.15: Weibull quantile

$F(x_p) = p$ gives

$$1 - e^{-\lambda x_p^\alpha} = p \implies e^{-\lambda x_p^\alpha} = 1 - p \implies \lambda x_p^\alpha = -\ln(1-p) \implies x_p = \Big[\frac{-\ln(1-p)}{\lambda}\Big]^{1/\alpha}.$$

Median: $(\ln 2/\lambda)^{1/\alpha}$.

Example 2.16: LogNormal quantile

$Y = \ln X \sim N(\mu, \sigma^2)$, and ln is increasing:

$$p = P(X \le x_p) = P(Y \le \ln x_p) = \Phi\Big(\frac{\ln x_p - \mu}{\sigma}\Big) \implies x_p = \exp\left(\mu + \sigma\Phi^{-1}(p)\right),$$

so $\mathrm{median}(X) = e^\mu$.

8 Median Residual Life

Definition 2.17: Median residual life

Preferred to the mrl for skewed distributions:

$$\operatorname{mdrl}(x) = \operatorname{median}(X - x \mid X > x),$$

equivalently

$$P(X - x > \operatorname{mdrl}(x) \mid X > x) = \frac{1}{2} \iff \frac{P(X > x + \operatorname{mdrl}(x),\, X > x)}{P(X > x)} = \frac{1}{2}.$$

For continuous X the numerator is $S(x + \operatorname{mdrl}(x))$:

$$S(x + \operatorname{mdrl}(x)) = \tfrac{1}{2}S(x) \quad \Longrightarrow \quad \operatorname{mdrl}(x) = S^{-1}\Big(\frac{S(x)}{2}\Big) - x.$$

Example 2.18: Exponential(λ)

$S^{-1}(y) = -(1/\lambda)\ln y$, so

$$\operatorname{mdrl}(x) = -\frac{1}{\lambda}\ln\Big(\frac{e^{-\lambda x}}{2}\Big) - x = -\frac{1}{\lambda}(-\lambda x - \ln 2) - x = \frac{\ln 2}{\lambda}.$$

constant in x.

9 Weibull Mean Residual Life

Let $S(t) = \exp(-\lambda t^{\alpha})$, $\lambda, \alpha > 0$, and $m(t) = \mathrm{E}[T - t \mid T > t]$.

1. Derive $m(t)$. 2. Show $m(t)$ is constant when $\alpha = 1$. 3. For $\alpha > 1$, show $m(t)$ is decreasing and interpret.

Derivation: substitution $v = \lambda u^{\alpha}$

With $f(u) = \lambda\alpha u^{\alpha-1}e^{-\lambda u^{\alpha}}$, the numerator is

$$N(t) = \int_t^{\infty} (u - t)\,\lambda\alpha u^{\alpha-1}e^{-\lambda u^{\alpha}}\,\mathrm{d}u.$$

With $v = \lambda u^{\alpha}$, $\mathrm{d}v = \lambda\alpha u^{\alpha-1}\,\mathrm{d}u$, $u = (v/\lambda)^{1/\alpha}$, limits λt^{α} to ∞:

$$\begin{aligned} N(t) &= \int_{\lambda t^{\alpha}}^{\infty} \Big(\big(\tfrac{v}{\lambda}\big)^{1/\alpha} - t\Big)e^{-v}\,\mathrm{d}v = \lambda^{-1/\alpha}\int_{\lambda t^{\alpha}}^{\infty} v^{1/\alpha}e^{-v}\,\mathrm{d}v - t\int_{\lambda t^{\alpha}}^{\infty} e^{-v}\,\mathrm{d}v \\ &= \lambda^{-1/\alpha}\,\Gamma\big(1 + \tfrac{1}{\alpha}, \lambda t^{\alpha}\big) - t\,e^{-\lambda t^{\alpha}}, \end{aligned}$$

with $\Gamma(s, x) = \int_x^{\infty} v^{s-1}e^{-v}\,\mathrm{d}v$ the upper incomplete gamma function. Dividing by $S(t) = e^{-\lambda t^{\alpha}}$:

$$m(t) = \frac{N(t)}{S(t)} = \frac{1}{S(t)} \cdot \frac{1}{\lambda^{1/\alpha}}\,\Gamma\big(1 + \tfrac{1}{\alpha}, \lambda t^{\alpha}\big) - t.$$

- $\alpha = 1$: $\Gamma(2, \lambda t) = (1 + \lambda t)e^{-\lambda t}$, so $m(t) = \frac{(1+\lambda t)}{\lambda} - t = \frac{1}{\lambda}$ (constant).
- $\alpha > 1$: $h(t) = \lambda\alpha t^{\alpha-1}$ increases, so $m(t)$ decreases (§10): the system *ages*.

10 mrl and Hazard Monotonicity

Motivation. Increasing risk means shrinking expected remaining life.

Theorem 2.19

Let $T \geq 0$ have survival S, density f, hazard $h = f/S$, and $m(t) = \mathrm{E}[T - t \mid T > t]$. Then

1. $m'(t) = h(t)m(t) - 1$;
2. if h is increasing then m is decreasing; if h is decreasing then m is increasing.

(1) Differential equation [Supplement]. $m(t) = \int_t^\infty S(u)\,\mathrm{d}u / S(t)$; by the quotient rule,

$$m'(t) = \frac{-S(t) \cdot S(t) - \big(\int_t^\infty S(u)\,\mathrm{d}u\big) S'(t)}{S(t)^2}.$$

With $S'(t) = -h(t)S(t)$,

$$m'(t) = \frac{-S(t)^2 + \big(\int_t^\infty S(u)\,\mathrm{d}u\big) h(t) S(t)}{S(t)^2} = -1 + h(t)\frac{\int_t^\infty S(u)\,\mathrm{d}u}{S(t)} = -1 + h(t)m(t).$$

■

(2) Monotonicity [Supplement]. By (1), $m'(t) \leq 0 \iff h(t)m(t) \leq 1$. Substituting $u \mapsto t + u$,

$$m(t) = \int_0^\infty \frac{S(t+u)}{S(t)}\,\mathrm{d}u = \int_0^\infty \exp\Big(-\int_t^{t+u} h(s)\,\mathrm{d}s\Big)\,\mathrm{d}u.$$

- h increasing: $\int_t^{t+u} h(s)\,\mathrm{d}s \geq u\,h(t)$, so

$$m(t) \leq \int_0^\infty e^{-uh(t)}\,\mathrm{d}u = \frac{1}{h(t)} \;\Rightarrow\; h(t)m(t) \leq 1 \;\Rightarrow\; m'(t) \leq 0.$$

- h decreasing: $\int_t^{t+u} h \leq u\,h(t)$, so $m(t) \geq 1/h(t)$ and $m'(t) \geq 0$.

■

Example 2.20: Application to the Weibull

$h(t) = \lambda\alpha t^{\alpha-1}$.

- $\alpha < 1$: h decreasing $\Rightarrow$ m increasing.
- $\alpha = 1$: $m'(t) = \lambda m(t) - 1$, bounded solution $m \equiv 1/\lambda$.
- $\alpha > 1$: h increasing $\Rightarrow$ m decreasing.

11 Selected exercises

Motivation. K&M Section 2.8: a discrete constant hazard, a mixture with falling hazard, and a distribution recovered from its mrl.

Exercise 2.1 (K&M 2.13: geometric distribution)

Let $P(X = x) = p(1-p)^{x-1}$, $x = 1, 2, \ldots$, with $0 < p < 1$.
(a) Find the survival function of X.
(b) Find the discrete hazard of X and compare it with the exponential hazard.

Solution.

(a) For integer $k \geq 0$,

$$P(X > k) = \sum_{j=k+1}^{\infty} p(1-p)^{j-1} = p(1-p)^k \sum_{i=0}^{\infty} (1-p)^i = (1-p)^k.$$

So $S(x) = (1-p)^{\lfloor x \rfloor}$, $x \geq 0$.

(b) For $j \geq 1$,

$$h(j) = \frac{P(X = j)}{P(X \geq j)} = \frac{p(1-p)^{j-1}}{S(j-1)} = \frac{p(1-p)^{j-1}}{(1-p)^{j-1}} = p,$$

constant, as for the exponential ($P(X > j + k \mid X > j) = (1-p)^k$). At integers $S(j) = e^{-\lambda j}$ with $\lambda = -\log(1-p) > p$: the Cox–Oakes cumulative hazard $j\lambda$, whereas K&M gives jp and $e^{-jp} \neq S(j)$. ■

Exercise 2.2 (K&M 2.14: gamma mixture of exponentials)

Given $\Theta = \theta$, the lifetime X is exponential with hazard θ; the individual hazard Θ has the gamma density $\lambda^\beta \theta^{\beta-1} e^{-\lambda\theta}/\Gamma(\beta)$, $\theta > 0$, with $\lambda, \beta > 0$.
(a) Find the survival function of X.
(b) Find the hazard rate of X and describe its shape.

Solution.

(a) $S(x) = \mathrm{E}[P(X > x \mid \Theta)] = \mathrm{E}[e^{-\Theta x}]$, and

$$\mathrm{E}[e^{-\Theta x}] = \int_0^\infty \frac{\lambda^\beta \theta^{\beta-1} e^{-(\lambda+x)\theta}}{\Gamma(\beta)}\, \mathrm{d}\theta = \frac{\lambda^\beta}{(\lambda+x)^\beta} \int_0^\infty \frac{(\lambda+x)^\beta \theta^{\beta-1} e^{-(\lambda+x)\theta}}{\Gamma(\beta)}\, \mathrm{d}\theta = \Big(\frac{\lambda}{\lambda+x}\Big)^\beta,$$

(the last integrand is the Gamma$(\lambda + x, \beta)$ density).

(b) $H(x) = \beta\{\log(\lambda + x) - \log\lambda\}$, so

$$h(x) = H'(x) = \frac{\beta}{\lambda + x},$$

strictly decreasing from $\beta/\lambda = \mathrm{E}(\Theta)$ to 0: individual hazards are constant, but the frail (large Θ) fail first. Indeed $\Theta \mid X > x$ has density $\propto e^{-\theta x}\theta^{\beta-1}e^{-\lambda\theta}$, Gamma$(\lambda + x, \beta)$, and $h(x) = \mathrm{E}(\Theta \mid X > x)$. (The marginal is Lomax.) ■

Exercise 2.3 (K&M 2.17: distribution from the mean residual life)

A continuous lifetime X has mean residual life $\mathrm{mrl}(x) = x + 10$ for all $x \geq 0$.
(a) Find $\mathrm{E}(X)$.
(b) Find $h(x)$.
(c) Find $S(x)$.

Solution.

(a) $\mathrm{E}(X) = \mathrm{mrl}(0) = 10$.

(b)

Step 1. With $R(x) = \int_x^\infty S(u)\,\mathrm{d}u$, the mrl needs $S > 0$, and $R(x) = (x+10)S(x)$, $x \geq 0$.

Step 2. R is continuous, so is $S = R/(x+10)$; then $R' = -S$ exists and S is differentiable.

Step 3. Differentiating, $-S(x) = S(x) + (x+10)S'(x)$, so

$$h(x) = -\frac{S'(x)}{S(x)} = \frac{2}{x+10}.$$

(Or by §10: $1 = m'(x) = h(x)m(x) - 1$.)

(c) $H(x) = \int_0^x \frac{2}{u+10}\,\mathrm{d}u = 2\log\frac{x+10}{10}$, so

$$S(x) = e^{-H(x)} = \Big(\frac{10}{x+10}\Big)^2, \qquad x \geq 0.$$

Check: $\int_x^\infty 100(u+10)^{-2}\,\mathrm{d}u = (x+10)S(x)$. Increasing mrl, decreasing hazard (§10). ■